Computations Fluid Dynamics (CFD) and Thermal Software
STATEMENT OF WORK (SOW)

1. Scope

This SOW covers Computation Fluid Dynamics (CFD) and Thermal Software Licenses, support, and maintenance over a five-year period. Software licenses, support, and maintenance services will be identified on individual Delivery Orders (DO).

2. Applicable Documents

Not applicable.

3. Technical Requirements

The contractor shall provide software licenses, support, and maintenance for the CFD and Thermal Software as follows:

<u>3.1 Software Requirements:</u>

<u>3.1.1 CFD Software shall be capable of the following:</u>

- Operate on both Windows and Linux systems
- Client-server architecture, ability to run simulations on a remote cluster via a local workstation with a user interface
- Incorporation of geometry modeling, pre-processing, meshing, solving, post-processing, and visualization features into a single unified software program
- Real-time monitoring and visualization of running simulations
- Graphical user interface (GUI) for user inputs to all aspects of geometry modeling, pre-processing, meshing, solving, post-processing, visualization.
- Geometry generation, modeling, and manipulations to include:
    - Ability to import varying industry standard three-dimensional (3D) geometries (Standard for the Exchange of Product Model Data (STEP), Initial Graphics Exchange Specification (IGES), Parasolid, Creo, Solidworks)
    - Ability to create original geometries
    - Ability to visualize and assess geometry pedigree
    - Ability to modify geometry
    - Ability to perform geometric Boolean of 3D geometries
- CFD Meshing capabilities to include
    - Two-dimensional (2D), Axisymmetric, and 3D meshing options
    - Automatic and manual mesh generations
    - Unstructured polyhedral mesh topology with prism later mesh
    - Unstructured trimmed hexahedral mesh topology with prism layer mesh
    - Overset unstructured mesh
    - Solution-based mesh refinement
    - Mesh morphing

    - Parallelized meshing
- Solver capabilities to include
    - Incompressible and compressible flow
    - Steady and unsteady
    - Internal and external flow
    - Single and multiple species
    - Reaching and non-reacting
    - Inviscid and laminar
    - Turbulence models (Menter Sheer Stress Transport (SST), Spalart-Allmaras, Large Eddy Simulation (LES), and Detached Eddy Simulation (DES)
    - Thermal non-equilibrium
    - Particle-laden flow
    - Natural convection
    - Conjugate heat transfer
    - Fluid-structure interaction
    - Rigid body motion
    - Rotating reference frame
    - Parallelized solver
- Optimization/Goal Seeking capabilities
- Post processing capabilities to include:
    - Reports and monitoring (forces, moments, mass flow, temperature, etc.)
    - Contour, Vector, and streamline plots
    - Visualization on cut-planes and surfaces
    - Line integral convolution plots
    - Histogram, XY plots, 3D plots
    - Export of visuals and tabular data
- User controls capabilities to include:
    - Custom field functions
    - Integration of custom scripting to automate tasks

3.1.2 Finite Element (FE) Thermal software shall be capable of:
- Operate on Windows systems
- Incorporation of geometry modeling, pre-processing, meshing, solving, post-processing, and visualization features into a single software suite.
- GUI for user inputs to all aspects of geometry modeling, pre-processing, meshing, solving, post-processing, visualization.
- Geometry generation, modeling, and manipulation to include:
    - Ability to import varying industry standard 3D geometries (STEP, IGES, Parasolid, Creo, Solidworks)
    - Ability to create original geometries
    - Ability to visualize and assess geometry pedigree
    - Ability to modify geometry
- FE Meshing capabilities to include:
    - 0D, 1D, 2D, Axisymmetric, and 3D meshing options
    - Automatic and manual mesh generations
    - Unstructured and structured mesh generations

- Solver Capabilities to include
  - Conduction, convection, and radiation heat transfer
  - Phase change thermal modeling
  - Radiation view factor calculation and ray tracing
  - Solar/space radiation
  - Diurnal heating
  - Steady state and transient analysis
- Post processing to include:
  - Reports and monitoring (temperature, heat flux, etc.)
  - Contour plots
  - Visualization on cut-planes and surfaces
  - Export of visual and tabular data
- User control capabilities to include
  - Custom field functions and tables
  - Integration of custom scripts to increase capabilities.

<u>3.1.3 Thermal system modeling software shall be capable of the following:</u>

- Operate on Windows systems
- Create virtual representations of complex, aerodynamic, mechanical, fluid, and thermal systems
- Pro and post processing visualization capabilities
- Integrates and couples with other Computer-Aided Design (CAD), Computer-Aided Engineering (CAE), and simulations software
- Aerodynamic system modeling capabilities include:
  - Handling of multi-physics aspects of flight control systems
  - Input aerodynamic properties from various source
  - Couples with mechanical, hydraulic, and thermal systems within Amesim
- Thermal system modeling capabilities include:
  - Simulating solid, liquid, and gas masses
  - Simulates phase change phenomena
  - Simulates conduction, convection, and radiation
  - Couples with mechanical and electrical systems within Amesim

Software renewal shall align with the existing yearly license renewal cycle.

3.2 Software Support and Maintenance

During the period of performance, the contractor shall provide technical services to troubleshoot installation and incorporation of the software on Navy Research Development Test and Evaluation (RDT&E) systems, providing support to local System Administrators (SA) and Information System Security Officers (ISSOs) via phone, virtual, or e-mail correspondence. The contractor shall provide engineering support to D555100 to troubleshoot software modeling issues via phone, virtual, or e-mail correspondence.

4. **Receipt, Acceptance, or Installation**

   The contractor shall provide updated license files compatible with RDT&E systems prior to each year's period of performance expiration.

   During the period of performance the contractor shall provide up-to-date software installation files for Windows and Linux via online website, SharePoint, e-mail, or other secure digital transfer methods.

5. **Security Requirements**

In the case of an unclassified contract that requires no access to Classified Military Information (CMI), but that does require access to Controlled Unclassified Information (CUI), NAWCWD Restricted Areas, and/or Government Information Technology (IT) systems, contractors are required to comply with non- National Industrial Security Program processing requirements before local access to CUI, Restricted Areas, and IT can be granted. This process shall be coordinated by the Contracting Officer's Representative (COR) or Technical Point of Contact (TPOC) through the NAWCWD Industrial Security office.

CUI information generated and/or provided under this contract shall be marked and safeguarded as specified in Department of Defense Instruction (DoDI) 5200.48, CUI. Contractor shall not store or transmit CUI on personal information technology systems or via personal e-mail. Unclassified e-mail containing any Department of Defense (DoD) CUI shall be encrypted. Prior to sending CUI to any non-Navy Marine Corps Internet (NMCI) addressees, the sender must first positively verify all recipients are authorized access to CUI and have need-to-know. Non-NMCI recipients must have a DoD compliant Private Key Infrastructure (PKI) certificate that enables electronic transmission via unclassified networks while protecting the CUI with a digital signature and encryption.

Public Release. Disclosure of information is covered by DFARS 252.204-7000 Disclosure of Information, incorporated in Section I of the contract. Concerning subsection (a)(2), "information otherwise in the public domain" is information officially released into the public domain, e.g. via Distribution Statement A, and does not include information in the public domain that has not been officially released. For disclosure of unclassified information that has not been officially released, the contractor must seek specific approval from the Contracting Officer, with approval from the NAWCWD Public Affairs Office (PAO).